

CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ SOLUTIONS

Chapter 19: Constructions (Circles)

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Circumscribed circle of a triangle

1. Using ruler and compasses only, construct $\triangle ABC$ with $AB = 6$ cm, $AC = 4.5$ cm and $\angle BAC = 120^\circ$. Construct the circle that passes through A , B and C , and measure its radius.

ICSE 2025, 4 marks

SOLUTION

Draw $AB = 6$ cm. At A construct an angle of 120° (two 60° arcs) and cut off $AC = 4.5$ cm on its arm. Join BC .

Draw the perpendicular bisectors of AB and AC . Let them meet at O .

With centre O and radius OA , draw the circle. It passes through A , B and C .

Because $\angle A$ is obtuse, O lies outside the triangle, on the other side of BC from A .

Check by calculation: $BC^2 = 6^2 + 4.5^2 - 2(6)(4.5) \cos 120^\circ = 83.25$, so $BC \approx 9.12$ cm and the radius is $\frac{BC}{2 \sin 120^\circ} \approx 5.27$ cm.

Answer: the radius measures about 5.3 cm

2. Using ruler and compasses, construct $\triangle ABC$ with $AB = 6$ cm, $\angle BAC = 120^\circ$ and $AC = 5$ cm. Construct the circle passing through A , B and C , and measure its radius. ICSE 2023, 4 marks

SOLUTION

Draw $AB = 6$ cm. At A construct an angle of 120° and cut off $AC = 5$ cm. Join BC .

Draw the perpendicular bisectors of AB and AC , meeting at O . With centre O and radius OA , draw the circle.

Check by calculation: $BC^2 = 6^2 + 5^2 - 2(6)(5) \cos 120^\circ = 91$, so $BC \approx 9.54$ cm and the radius is $\frac{BC}{2 \sin 120^\circ} \approx 5.5$ cm.

Answer: the radius measures about 5.5 cm

3. Using ruler and compasses, construct $\triangle ABC$ with $AB = 3$ cm, $BC = 4$ cm and $\angle ABC = 90^\circ$. Construct its circumscribing circle and measure the radius. **ICSE 2020, 4 marks**

SOLUTION

Draw $BC = 4$ cm, construct a right angle at B and cut off $BA = 3$ cm. Join AC .

Draw the perpendicular bisectors of AB and BC , meeting at O . With centre O and radius OA , draw the circle.

Check: in a right-angled triangle the circumcentre is the mid-point of the hypotenuse. $AC = \sqrt{3^2 + 4^2} = 5$ cm, so the radius is 2.5 cm.

Answer: radius = 2.5 cm

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